Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

April 01, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids-Spring
Pace Project No.: 50396428

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 18, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

CERTIFICATIONS

Project:	Biosolids-Spring
Pace Project No.:	50396428

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268	Michigan Drinking Water Laboratory #9050
Illinois Accreditation #: 200074	Oklahoma Laboratory #: 9204
Indiana Drinking Water Laboratory #: C-49-06	Texas Certification #: T104704355
Kansas/TNI Certification #: E-10177	Washington Dept of Ecology #: C1081
Kentucky UST Agency Interest #: 80226	Wisconsin Laboratory #: 999788130
Kentucky WW Laboratory ID #: 98019	USDA Foreign Soil Permit #: 525-23-13-23119
Louisiana Certification #: 04076	USDA Compliance Agreement #: IN-SL-22-001

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SAMPLE SUMMARY

Project: Biosolids-Spring
Pace Project No.: 50396428

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50396428001	**DIGESTER #3**	Solid	03/17/25 13:14	03/18/25 10:25

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SAMPLE ANALYTE COUNT

Project: Biosolids-Spring
Pace Project No.: 50396428

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50396428001	**DIGESTER #3**	EPA 8081	ADM	21	PASI-I
		EPA 8081	KAV	8	PASI-I
		EPA 8082	DMP1	8	PASI-I
		EPA 8151A	BJW	3	PASI-I
		EPA 6010	FAC	6	PASI-I
		EPA 6010	ELK	8	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7470	EEM	1	PASI-I
		EPA 7471	MGM	1	PASI-I
		EPA 8270	FIP	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	SLB	55	PASI-I
		EPA 5030/8260	TMW	13	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		EPA 9038	AEL	1	PASI-I
		EPA 9045	JDR	1	PASI-I
		EPA 351.2	STS	1	PASI-I
		EPA 353.2	KCS	2	PASI-I
		SM 4500-NH3 G	STS	1	PASI-I
		SM 4500-Cl-E	KCS	1	PASI-I
		SM 4500-CN-E	JTR	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids	Analytical Method: EPA 8081 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Aldrin	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	309-00-2	
alpha-BHC	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	319-84-6	P1
beta-BHC	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	319-85-7	
delta-BHC	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	319-86-8	
gamma-BHC (Lindane)	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	58-89-9	
Chlordane (Technical)	**<32200**	ug/kg	32200	1	03/28/25 10:45	03/28/25 18:33	57-74-9	
alpha-Chlordane	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	5103-71-9	
gamma-Chlordane	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	5103-74-2	
4,4'-DDD	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	72-54-8	
4,4'-DDE	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	72-55-9	
4,4'-DDT	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	50-29-3	
Dieldrin	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	60-57-1	
Endosulfan I	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	959-98-8	
Endosulfan II	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	33213-65-9	
Endosulfan sulfate	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	1031-07-8	
Endrin	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	72-20-8	
Endrin aldehyde	**<3220**	ug/kg	3220	1	03/28/25 10:45	03/28/25 18:33	7421-93-4	
Heptachlor	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	76-44-8	
Heptachlor epoxide	**<1610**	ug/kg	1610	1	03/28/25 10:45	03/28/25 18:33	1024-57-3	
Toxaphene	**<32200**	ug/kg	32200	1	03/28/25 10:45	03/28/25 18:33	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	74	%.	10-139	1	03/28/25 10:45	03/28/25 18:33	2051-24-3	
8081 GCS Pest RV, TCLP	Analytical Method: EPA 8081 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16							
	Pace Analytical Services - Indianapolis							
gamma-BHC (Lindane)	**<0.00025**	mg/L	0.00025	1	03/22/25 19:40	03/24/25 14:00	58-89-9	
Chlordane (Technical)	**<0.0050**	mg/L	0.0050	1	03/22/25 19:40	03/24/25 14:00	57-74-9	
Endrin	**<0.00050**	mg/L	0.00050	1	03/22/25 19:40	03/24/25 14:00	72-20-8	
Heptachlor	**<0.00025**	mg/L	0.00025	1	03/22/25 19:40	03/24/25 14:00	76-44-8	
Heptachlor epoxide	**<0.00025**	mg/L	0.00025	1	03/22/25 19:40	03/24/25 14:00	1024-57-3	
Methoxychlor	**<0.0025**	mg/L	0.0025	1	03/22/25 19:40	03/24/25 14:00	72-43-5	
Toxaphene	**<0.0050**	mg/L	0.0050	1	03/22/25 19:40	03/24/25 14:00	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	53	%.	10-128	1	03/22/25 19:40	03/24/25 14:00	2051-24-3	
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
PCB-1016 (Aroclor 1016)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	12674-11-2	
PCB-1221 (Aroclor 1221)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	11104-28-2	
PCB-1232 (Aroclor 1232)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	11141-16-5	
PCB-1242 (Aroclor 1242)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	53469-21-9	
PCB-1248 (Aroclor 1248)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	12672-29-6	
PCB-1254 (Aroclor 1254)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	11097-69-1	
PCB-1260 (Aroclor 1260)	**<3220**	ug/kg	3220	1	03/22/25 16:11	03/28/25 23:05	11096-82-5	

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Surrogates								
Tetrachloro-m-xylene (S)	57	%.	11-126	1	03/22/25 16:11	03/28/25 23:05	877-09-8	
8151A Cl Acid Herbicides TCLP	Analytical Method: EPA 8151A Preparation Method: EPA 8151							
	Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16							
	Pace Analytical Services - Indianapolis							
2,4-D	**<0.0050**	mg/L	0.0050	1	03/24/25 15:04	03/28/25 16:38	94-75-7	
2,4,5-TP (Silvex)	**<0.0050**	mg/L	0.0050	1	03/24/25 15:04	03/28/25 16:38	93-72-1	
Surrogates								
2,4-DCAA (S)	67	%.	30-170	1	03/24/25 15:04	03/28/25 16:38	19719-28-9	
6010 MET ICP	Analytical Method: EPA 6010 Preparation Method: EPA 3050							
	Pace Analytical Services - Indianapolis							
Aluminum	**4420**	mg/kg	1360	1	03/25/25 15:49	03/26/25 11:40	7429-90-5	
Calcium	**28300**	mg/kg	1360	1	03/25/25 15:49	03/27/25 14:19	7440-70-2	
Magnesium	**4940**	mg/kg	1360	1	03/25/25 15:49	03/26/25 11:40	7439-95-4	
Phosphorus	**39400**	mg/kg	1360	1	03/25/25 15:49	03/26/25 11:40	7723-14-0	N2
Potassium	**2840**	mg/kg	1360	1	03/25/25 15:49	03/26/25 11:40	7440-09-7	
Sodium	**4810**	mg/kg	1360	1	03/25/25 15:49	03/26/25 11:40	7440-23-5	
6010 MET ICP, TCLP	Analytical Method: EPA 6010 Preparation Method: EPA 3010							
	Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16							
	Pace Analytical Services - Indianapolis							
Arsenic	**<0.10**	mg/L	0.10	1	03/24/25 15:57	03/27/25 15:17	7440-38-2	
Barium	**<5.0**	mg/L	5.0	1	03/24/25 15:57	03/27/25 15:17	7440-39-3	
Cadmium	**<0.050**	mg/L	0.050	1	03/24/25 15:57	03/27/25 15:17	7440-43-9	
Chromium	**<0.10**	mg/L	0.10	1	03/24/25 15:57	03/27/25 15:17	7440-47-3	
Lead	**<0.10**	mg/L	0.10	1	03/24/25 15:57	03/27/25 15:17	7439-92-1	
Selenium	**<0.10**	mg/L	0.10	1	03/24/25 15:57	03/27/25 15:17	7782-49-2	
Silver	**<0.10**	mg/L	0.10	1	03/24/25 15:57	03/27/25 15:17	7440-22-4	
Zinc	**1.9**	mg/L	0.20	1	03/24/25 15:57	03/27/25 15:17	7440-66-6	
6020 MET ICPMS	Analytical Method: EPA 6020 Preparation Method: EPA 3050B							
	Pace Analytical Services - Indianapolis							
Arsenic	**18.1**	mg/kg	3.1	1	03/26/25 12:28	03/27/25 04:30	7440-38-2	
Cadmium	**2.6**	mg/kg	1.6	1	03/26/25 12:28	03/27/25 04:30	7440-43-9	
Chromium	**115**	mg/kg	6.2	1	03/26/25 12:28	03/27/25 04:30	7440-47-3	
Copper	**827**	mg/kg	6.2	2	03/26/25 12:28	03/27/25 04:44	7440-50-8	
Lead	**15.5**	mg/kg	3.1	1	03/26/25 12:28	03/27/25 04:30	7439-92-1	
Molybdenum	**19.8**	mg/kg	3.1	1	03/26/25 12:28	03/27/25 04:30	7439-98-7	N2
Nickel	**23.7**	mg/kg	3.1	1	03/26/25 12:28	03/27/25 04:30	7440-02-0	
Selenium	**6.1**	mg/kg	3.1	1	03/26/25 12:28	03/27/25 04:30	7782-49-2	
Silver	**2.9**	mg/kg	1.6	1	03/26/25 12:28	03/27/25 04:30	7440-22-4	
Zinc	**1100**	mg/kg	31.2	2	03/26/25 12:28	03/27/25 04:44	7440-66-6	

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP	Analytical Method: EPA 7470 Preparation Method: EPA 7470 Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16 Pace Analytical Services - Indianapolis							
Mercury	**<0.0020**	mg/L	0.0020	1	03/21/25 11:10	03/23/25 21:32	7439-97-6	
7471 Mercury	Analytical Method: EPA 7471 Preparation Method: EPA 7471 Pace Analytical Services - Indianapolis							
Mercury	**<6.2**	mg/kg	6.2	1	03/30/25 20:10	03/31/25 08:44	7439-97-6	
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis							
Acenaphthene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	83-32-9	
Acenaphthylene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	208-96-8	
Anthracene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	120-12-7	
Benzo(a)anthracene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	56-55-3	
Benzo(a)pyrene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	50-32-8	
Benzo(b)fluoranthene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	205-99-2	
Benzo(g,h,i)perylene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	191-24-2	
Benzo(k)fluoranthene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	207-08-9	
Benzyl alcohol	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48	100-51-6	
4-Bromophenylphenyl ether	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	101-55-3	
Butylbenzylphthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	85-68-7	
4-Chloro-3-methylphenol	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48	59-50-7	
4-Chloroaniline	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48	106-47-8	
bis(2-Chloroethoxy)methane	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	111-91-1	
bis(2-Chloroethyl) ether	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	111-44-4	
bis(2chloro1methylethyl) ether	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	108-60-1	
2-Chloronaphthalene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	91-58-7	
2-Chlorophenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	95-57-8	
4-Chlorophenylphenyl ether	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	7005-72-3	
Chrysene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	218-01-9	
Dibenz(a,h)anthracene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	53-70-3	
Dibenzofuran	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	132-64-9	
3,3'-Dichlorobenzidine	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48	91-94-1	
2,4-Dichlorophenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	120-83-2	
Diethylphthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	84-66-2	
2,4-Dimethylphenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	105-67-9	
Dimethylphthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	131-11-3	
Di-n-butylphthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	84-74-2	
4,6-Dinitro-2-methylphenol	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48	534-52-1	
2,4-Dinitrophenol	**<1100000**	ug/kg	1100000	1	03/24/25 14:54	03/25/25 15:48	51-28-5	
2,4-Dinitrotoluene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	121-14-2	
2,6-Dinitrotoluene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	606-20-2	
Di-n-octylphthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	117-84-0	
bis(2-Ethylhexyl)phthalate	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	117-81-7	
Fluoranthene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	206-44-0	

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Fluorene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	86-73-7	
Hexachloro-1,3-butadiene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	87-68-3	
Hexachlorobenzene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	118-74-1	
Hexachlorocyclopentadiene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	77-47-4	
Hexachloroethane	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	67-72-1	
Indeno(1,2,3-cd)pyrene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	193-39-5	
Isophorone	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	78-59-1	
1-Methylnaphthalene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	90-12-0	
2-Methylnaphthalene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	91-57-6	
2-Methylphenol(o-Cresol)	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<455000**	ug/kg	455000	1	03/24/25 14:54	03/25/25 15:48		
Naphthalene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	91-20-3	
2-Nitroaniline	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	88-74-4	
3-Nitroaniline	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	99-09-2	
4-Nitroaniline	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	100-01-6	
Nitrobenzene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	98-95-3	
2-Nitrophenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	88-75-5	
4-Nitrophenol	**<1100000**	ug/kg	1100000	1	03/24/25 14:54	03/25/25 15:48	100-02-7	
N-Nitroso-di-n-propylamine	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	621-64-7	
N-Nitrosodiphenylamine	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	86-30-6	
Pentachlorophenol	**<1100000**	ug/kg	1100000	1	03/24/25 14:54	03/25/25 15:48	87-86-5	
Phenanthrene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	85-01-8	
Phenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	108-95-2	P1
Pyrene	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	129-00-0	
2,4,5-Trichlorophenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	95-95-4	
2,4,6-Trichlorophenol	**<228000**	ug/kg	228000	1	03/24/25 14:54	03/25/25 15:48	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	70	%.	32-98	1	03/24/25 14:54	03/25/25 15:48	4165-60-0	
Phenol-d5 (S)	87	%.	30-115	1	03/24/25 14:54	03/25/25 15:48	4165-62-2	
2-Fluorophenol (S)	81	%.	18-116	1	03/24/25 14:54	03/25/25 15:48	367-12-4	
2,4,6-Tribromophenol (S)	87	%.	15-140	1	03/24/25 14:54	03/25/25 15:48	118-79-6	
2-Fluorobiphenyl (S)	72	%.	39-97	1	03/24/25 14:54	03/25/25 15:48	321-60-8	
p-Terphenyl-d14 (S)	90	%.	49-119	1	03/24/25 14:54	03/25/25 15:48	1718-51-0	
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16							
	Pace Analytical Services - Indianapolis							
1,4-Dichlorobenzene	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	106-46-7	
2,4-Dinitrotoluene	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	121-14-2	
Hexachloro-1,3-butadiene	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	87-68-3	
Hexachlorobenzene	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	118-74-1	
Hexachloroethane	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	67-72-1	
2-Methylphenol(o-Cresol)	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<0.20**	mg/L	0.20	1	03/26/25 16:44	03/29/25 22:26		
Nitrobenzene	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	98-95-3	

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 03/20/25 13:40 Initial pH: 8.58; Final pH: 5.16							
	Pace Analytical Services - Indianapolis							
Pentachlorophenol	**<0.50**	mg/L	0.50	1	03/26/25 16:44	03/29/25 22:26	87-86-5	
Pyridine	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	110-86-1	
2,4,5-Trichlorophenol	**<0.50**	mg/L	0.50	1	03/26/25 16:44	03/29/25 22:26	95-95-4	
2,4,6-Trichlorophenol	**<0.10**	mg/L	0.10	1	03/26/25 16:44	03/29/25 22:26	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	62	%.	41-102	1	03/26/25 16:44	03/29/25 22:26	4165-60-0	
2-Fluorobiphenyl (S)	46	%.	29-84	1	03/26/25 16:44	03/29/25 22:26	321-60-8	
p-Terphenyl-d14 (S)	91	%.	54-119	1	03/26/25 16:44	03/29/25 22:26	1718-51-0	
Phenol-d5 (S)	36	%.	18-51	1	03/26/25 16:44	03/29/25 22:26	4165-62-2	
2-Fluorophenol (S)	48	%.	25-76	1	03/26/25 16:44	03/29/25 22:26	367-12-4	
2,4,6-Tribromophenol (S)	97	%.	49-147	1	03/26/25 16:44	03/29/25 22:26	118-79-6	
8260 MSV 5030 Low Level	Analytical Method: EPA 8260							
	Pace Analytical Services - Indianapolis							
Acetone	**6.8**	mg/kg	3.0	1		03/24/25 11:30	67-64-1	
Acrolein	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	107-02-8	
Acrylonitrile	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	107-13-1	
Benzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	71-43-2	
Bromodichloromethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-27-4	
Bromoform	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-25-2	
Bromomethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	74-83-9	
2-Butanone (MEK)	**1.3**	mg/kg	0.74	1		03/24/25 11:30	78-93-3	
Carbon disulfide	**<0.30**	mg/kg	0.30	1		03/24/25 11:30	75-15-0	
Carbon tetrachloride	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	56-23-5	
Chlorobenzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	108-90-7	
Chloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-00-3	
Chloroform	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	67-66-3	
Chloromethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	74-87-3	
1,2-Dibromo-3-chloropropane	**<0.30**	mg/kg	0.30	1		03/24/25 11:30	96-12-8	
Dibromochloromethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	124-48-1	
1,2-Dibromoethane (EDB)	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	106-93-4	
Dibromomethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	74-95-3	N2
1,2-Dichlorobenzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	95-50-1	
1,3-Dichlorobenzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	541-73-1	
1,4-Dichlorobenzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	106-46-7	
trans-1,4-Dichloro-2-butene	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	110-57-6	
Dichlorodifluoromethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-71-8	
1,1-Dichloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-34-3	
1,2-Dichloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	107-06-2	
1,1-Dichloroethene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-35-4	
cis-1,2-Dichloroethene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	156-59-2	
trans-1,2-Dichloroethene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	156-60-5	
1,2-Dichloropropane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	78-87-5	
cis-1,3-Dichloropropene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	10061-01-5	

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level	Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis							
trans-1,3-Dichloropropene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	10061-02-6	
Ethylbenzene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	100-41-4	
Ethyl methacrylate	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	97-63-2	
2-Hexanone	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	591-78-6	
Iodomethane	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	74-88-4	
Methylene Chloride	**<0.60**	mg/kg	0.60	1		03/24/25 11:30	75-09-2	
Methyl methacrylate	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	80-62-6	
4-Methyl-2-pentanone (MIBK)	**<0.74**	mg/kg	0.74	1		03/24/25 11:30	108-10-1	
Methyl-tert-butyl ether	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	1634-04-4	
Styrene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	100-42-5	
1,1,1,2-Tetrachloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	630-20-6	
1,1,2,2-Tetrachloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	79-34-5	
Tetrachloroethene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	127-18-4	
Toluene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	108-88-3	
1,1,1-Trichloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	71-55-6	
1,1,2-Trichloroethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	79-00-5	
Trichloroethene	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	79-01-6	
Trichlorofluoromethane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-69-4	
1,2,3-Trichloropropane	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	96-18-4	
Vinyl acetate	**<3.0**	mg/kg	3.0	1		03/24/25 11:30	108-05-4	
Vinyl chloride	**<0.15**	mg/kg	0.15	1		03/24/25 11:30	75-01-4	
Xylene (Total)	**<0.30**	mg/kg	0.30	1		03/24/25 11:30	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	105	%.	75-135	1		03/24/25 11:30	1868-53-7	
Toluene-d8 (S)	107	%.	65-148	1		03/24/25 11:30	2037-26-5	
4-Bromofluorobenzene (S)	96	%.	63-132	1		03/24/25 11:30	460-00-4	
8260 MSV TCLP	Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 03/20/25 13:40 Pace Analytical Services - Indianapolis							
Benzene	**<0.050**	mg/L	0.050	1		03/21/25 20:57	71-43-2	
2-Butanone (MEK)	**<1.0**	mg/L	1.0	1		03/21/25 20:57	78-93-3	
Carbon tetrachloride	**<0.050**	mg/L	0.050	1		03/21/25 20:57	56-23-5	
Chlorobenzene	**<0.050**	mg/L	0.050	1		03/21/25 20:57	108-90-7	
Chloroform	**<0.050**	mg/L	0.050	1		03/21/25 20:57	67-66-3	
1,2-Dichloroethane	**<0.050**	mg/L	0.050	1		03/21/25 20:57	107-06-2	
1,1-Dichloroethene	**<0.050**	mg/L	0.050	1		03/21/25 20:57	75-35-4	
Tetrachloroethene	**<0.050**	mg/L	0.050	1		03/21/25 20:57	127-18-4	
Trichloroethene	**<0.050**	mg/L	0.050	1		03/21/25 20:57	79-01-6	
Vinyl chloride	**<0.020**	mg/L	0.020	1		03/21/25 20:57	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	98	%.	79-124	1		03/21/25 20:57	460-00-4	
Dibromofluoromethane (S)	108	%.	82-128	1		03/21/25 20:57	1868-53-7	
Toluene-d8 (S)	101	%.	73-122	1		03/21/25 20:57	2037-26-5	

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ANALYTICAL RESULTS

Project: Biosolids-Spring
Pace Project No.: 50396428

Sample: DIGESTER #3 **Lab ID: 50396428001** Collected: 03/17/25 13:14 Received: 03/18/25 10:25 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Percent Moisture	**96.9**	%	0.10	1		03/20/25 16:58		N2
2540G Total Percent Solids	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Total Solids	**3.1**	%	0.10	1		03/20/25 16:58		N2
9038 Sulfate Soil	Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis							
Sulfate	**6230**	mg/kg	3220	1	03/27/25 16:29	03/28/25 09:29	14808-79-8	N2
9045 pH Soil	Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis							
pH at 25 Degrees C	**8.4**	Std. Units	0.10	1		03/27/25 15:53		H3
351.2 Total Kjeldahl Nitrogen	Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis							
Nitrogen, Kjeldahl, Total	**69400**	mg/kg	7770	5	03/24/25 09:45	03/27/25 11:41	7727-37-9	N2
353.2 Nitrogen, NO2/NO3	Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis							
Nitrogen, NO2 plus NO3	**<1610**	mg/kg	1610	10	03/24/25 12:55	03/26/25 08:59		D3,N2
Nitrogen, Nitrate	**<1610**	mg/kg	1610	10	03/24/25 12:55	03/26/25 08:59	14797-55-8	D3,N2
4500 Ammonia Soil, Distilled	Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis							
Nitrogen, Ammonia	**17400**	mg/kg	1010	1	03/26/25 11:45	03/26/25 15:26	7664-41-7	N2
4500 Chloride in Soil	Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis							
Chloride	**<16100**	mg/kg	16100	5	03/24/25 12:35	03/25/25 10:31	16887-00-6	D3,N2
4500CNE Cyanide, Soil	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis							
Cyanide	**<15.6**	mg/kg	15.6	1	03/25/25 12:45	03/27/25 14:46	57-12-5	N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835468
QC Batch Method: EPA 7470
Analysis Method: EPA 7470
Analysis Description: 7470 Mercury TCLP
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3822839 Matrix: Water
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	03/23/25 20:51	

LABORATORY CONTROL SAMPLE: 3822840

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0048	97	80-120	

MATRIX SPIKE SAMPLE: 3822841

Parameter	Units	50396269001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	99	75-125	

MATRIX SPIKE SAMPLE: 3822842

Parameter	Units	50396270001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	100	75-125	

MATRIX SPIKE SAMPLE: 3822843

Parameter	Units	50396529001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.0029	19	75-125	M0

MATRIX SPIKE SAMPLE: 3822844

Parameter	Units	50396350001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	101	75-125	

MATRIX SPIKE SAMPLE: 3822845

Parameter	Units	50396423001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.015	97	75-125	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE SAMPLE: 3822846

Parameter	Units	50396428001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.015	101	75-125	

MATRIX SPIKE SAMPLE: 3822847

Parameter	Units	50396506001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	94	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3822848 3822849

Parameter	Units	50396637002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	<0.0020	0.015	0.015	0.017	0.016	98	95	75-125	3	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 836325
QC Batch Method: EPA 7471
Analysis Method: EPA 7471
Analysis Description: 7471 Mercury
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3827289 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	03/31/25 08:37	

LABORATORY CONTROL SAMPLE: 3827290

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.49	99	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3827291 3827292

Parameter	Units	50396648003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	1.8	0.55	0.55	1.7	4.0	-33	402	75-125	83	20	M0,R1

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835177
QC Batch Method: EPA 3050
Analysis Method: EPA 6010
Analysis Description: 6010 MET
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3821415 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	03/26/25 13:36	
Calcium	mg/kg	<50.0	50.0	03/26/25 13:36	
Magnesium	mg/kg	<50.0	50.0	03/26/25 13:36	
Phosphorus	mg/kg	ND	50.0	03/26/25 13:36	N2
Potassium	mg/kg	<50.0	50.0	03/26/25 13:36	
Sodium	mg/kg	<50.0	50.0	03/26/25 13:36	

LABORATORY CONTROL SAMPLE: 3821416

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	523	105	80-120	
Calcium	mg/kg	500	538	108	80-120	
Magnesium	mg/kg	500	521	104	80-120	
Phosphorus	mg/kg	50	<50.0	99	80-120	N2
Potassium	mg/kg	500	510	102	80-120	
Sodium	mg/kg	500	515	103	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3821417 3821418

Parameter	Units	50396557001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	11200	502	536	13600	8710	482	-461	75-125	44	20	P6,R1
Calcium	mg/kg	1900	502	536	1940	66200	8	12000	75-125	189	20	E,M0, R1
Magnesium	mg/kg	1940	502	536	2420	3410	96	274	75-125	34	20	M0,R1
Phosphorus	mg/kg	361	50.2	53.6	389	310	56	-96	75-125	23	20	N2,P6, R1
Potassium	mg/kg	888	502	536	1690	1430	159	101	75-125	17	20	M0
Sodium	mg/kg	208	502	536	624	671	83	86	75-125	7	20	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835549 — Analysis Method: EPA 6010
QC Batch Method: EPA 3010 — Analysis Description: 6010 MET TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3823532 — Matrix: Water

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	03/27/25 15:13	
Barium	mg/L	<0.50	0.50	03/27/25 15:13	
Cadmium	mg/L	<0.0050	0.0050	03/27/25 15:13	
Chromium	mg/L	<0.010	0.010	03/27/25 15:13	
Lead	mg/L	<0.010	0.010	03/27/25 15:13	
Selenium	mg/L	<0.010	0.010	03/27/25 15:13	
Silver	mg/L	<0.010	0.010	03/27/25 15:13	
Zinc	mg/L	<0.020	0.020	03/27/25 15:13	

LABORATORY CONTROL SAMPLE: 3823533

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	0.97	97	80-120	
Barium	mg/L	1	0.99	99	80-120	
Cadmium	mg/L	1	0.95	95	80-120	
Chromium	mg/L	1	0.99	99	80-120	
Lead	mg/L	1	0.92	92	80-120	
Selenium	mg/L	1	0.96	96	80-120	
Silver	mg/L	0.5	0.49	98	80-120	
Zinc	mg/L	1	0.98	98	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3823534 3823535

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	<0.10	10	10	9.8	9.8	98	98	50-150	0	20	
Barium	mg/L	<5.0	10	10	9.6	9.7	95	96	50-150	1	20	
Cadmium	mg/L	<0.050	10	10	9.3	9.2	93	92	50-150	0	20	
Chromium	mg/L	<0.10	10	10	9.5	9.6	95	96	50-150	1	20	
Lead	mg/L	<0.10	10	10	8.7	8.7	87	87	50-150	0	20	
Selenium	mg/L	<0.10	10	10	9.6	9.6	96	96	50-150	0	20	
Silver	mg/L	<0.10	5	5	4.7	4.7	94	95	50-150	1	20	
Zinc	mg/L	1.9	10	10	11.6	11.5	97	96	50-150	0	20	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE SAMPLE: 3823536

Parameter	Units	50396465001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.9	99	50-150	
Barium	mg/L	ND	10	10.6	97	50-150	
Cadmium	mg/L	ND	10	9.4	94	50-150	
Chromium	mg/L	ND	10	9.8	98	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	9.7	97	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	ND	10	9.8	98	50-150	

MATRIX SPIKE SAMPLE: 3823537

Parameter	Units	50396393001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.9	99	50-150	
Barium	mg/L	ND	10	11.3	95	50-150	
Cadmium	mg/L	ND	10	9.4	94	50-150	
Chromium	mg/L	0.38	10	9.9	95	50-150	
Lead	mg/L	ND	10	8.7	87	50-150	
Selenium	mg/L	ND	10	9.7	97	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	ND	10	9.6	96	50-150	

MATRIX SPIKE SAMPLE: 3823538

Parameter	Units	50396482001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.8	97	50-150	
Barium	mg/L	ND	10	10	95	50-150	
Cadmium	mg/L	ND	10	9.3	93	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	8.8	88	50-150	
Selenium	mg/L	ND	10	9.6	96	50-150	
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	ND	10	9.7	97	50-150	

MATRIX SPIKE SAMPLE: 3823539

Parameter	Units	50396506001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	0.20	10	10.1	99	50-150	
Barium	mg/L	ND	10	9.6	95	50-150	
Cadmium	mg/L	ND	10	9.2	92	50-150	
Chromium	mg/L	ND	10	9.5	95	50-150	
Lead	mg/L	ND	10	8.6	86	50-150	
Selenium	mg/L	ND	10	9.6	96	50-150	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE SAMPLE: 3823539

Parameter	Units	50396506001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	ND	10	9.8	97	50-150	

MATRIX SPIKE SAMPLE: 3823540

Parameter	Units	50396541002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	145 ug/L	10	9.5	94	50-150	
Barium	mg/L	ND	10	9.5	91	50-150	
Cadmium	mg/L	ND	10	8.9	89	50-150	
Chromium	mg/L	ND	10	9.2	92	50-150	
Lead	mg/L	ND	10	8.4	84	50-150	
Selenium	mg/L	ND	10	9.2	92	50-150	
Silver	mg/L	ND	5	4.6	92	50-150	
Zinc	mg/L	2590 ug/L	10	11.8	92	50-150	

MATRIX SPIKE SAMPLE: 3823541

Parameter	Units	50396581001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.7	97	50-150	
Barium	mg/L	ND	10	9.4	94	50-150	
Cadmium	mg/L	ND	10	9.2	92	50-150	
Chromium	mg/L	ND	10	9.5	95	50-150	
Lead	mg/L	ND	10	8.6	86	50-150	
Selenium	mg/L	ND	10	9.5	95	50-150	
Silver	mg/L	ND	5	4.7	93	50-150	
Zinc	mg/L	ND	10	9.6	96	50-150	

MATRIX SPIKE SAMPLE: 3823542

Parameter	Units	50396607001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.7	97	50-150	
Barium	mg/L	ND	10	9.7	96	50-150	
Cadmium	mg/L	ND	10	9.2	92	50-150	
Chromium	mg/L	ND	10	9.6	95	50-150	
Lead	mg/L	ND	10	8.7	87	50-150	
Selenium	mg/L	ND	10	9.5	95	50-150	
Silver	mg/L	ND	5	4.7	94	50-150	
Zinc	mg/L	879 ug/L	10	10.4	96	50-150	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE SAMPLE: 3823543

Parameter	Units	50396641001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10	99	50-150	
Barium	mg/L	ND	10	9.7	96	50-150	
Cadmium	mg/L	ND	10	9.5	94	50-150	
Chromium	mg/L	ND	10	9.7	97	50-150	
Lead	mg/L	ND	10	8.9	88	50-150	
Selenium	mg/L	ND	10	9.8	98	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	5360 ug/L	10	15.3	100	50-150	

MATRIX SPIKE SAMPLE: 3823544

Parameter	Units	50396637001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	0.39	10	10.1	97	50-150	
Barium	mg/L	<5.0	10	9.6	95	50-150	
Cadmium	mg/L	<0.050	10	9.3	93	50-150	
Chromium	mg/L	29.4	10	39.2	97	50-150	
Lead	mg/L	<0.10	10	8.7	87	50-150	
Selenium	mg/L	<0.10	10	9.5	95	50-150	
Silver	mg/L	<0.10	5	4.7	95	50-150	
Zinc	mg/L	3130 ug/L	10	12.8	97	50-150	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835811
QC Batch Method: EPA 3050B
Analysis Method: EPA 6020
Analysis Description: 6020 MET
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824814 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	03/27/25 04:15	
Cadmium	mg/kg	<0.050	0.050	03/27/25 04:15	
Chromium	mg/kg	<0.20	0.20	03/27/25 04:15	
Copper	mg/kg	<0.10	0.10	03/27/25 04:15	
Lead	mg/kg	<0.10	0.10	03/27/25 04:15	
Molybdenum	mg/kg	<0.10	0.10	03/27/25 04:15	N2
Nickel	mg/kg	<0.10	0.10	03/27/25 04:15	
Selenium	mg/kg	<0.10	0.10	03/27/25 04:15	
Silver	mg/kg	<0.050	0.050	03/27/25 04:15	
Zinc	mg/kg	<0.50	0.50	03/27/25 04:15	

LABORATORY CONTROL SAMPLE: 3824815

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.9	99	80-120	
Cadmium	mg/kg	4	4.2	104	80-120	
Chromium	mg/kg	4	4.1	103	80-120	
Copper	mg/kg	4	4.1	103	80-120	
Lead	mg/kg	4	4.1	103	80-120	
Molybdenum	mg/kg	4	4.2	104	80-120	N2
Nickel	mg/kg	4	4.1	102	80-120	
Selenium	mg/kg	4	4.0	100	80-120	
Silver	mg/kg	4	4.1	104	80-120	
Zinc	mg/kg	4	4.0	101	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824816 3824817

Parameter	Units	50396518004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	2.5	31.6	32.4	33.8	33.7	98	98	75-125	0	20	
Cadmium	mg/kg	1.4	31.6	32.4	35.1	34.5	106	103	75-125	2	20	
Chromium	mg/kg	7.6	31.6	32.4	39.9	40.6	101	103	75-125	2	20	
Copper	mg/kg	37.6	31.6	32.4	70.2	70.7	102	103	75-125	1	20	
Lead	mg/kg	6.1	31.6	32.4	39.1	38.9	103	102	75-125	1	20	
Molybdenum	mg/kg	2.9	31.6	32.4	35.1	35.1	101	101	75-125	0	20	N2
Nickel	mg/kg	5.3	31.6	32.4	36.9	36.6	99	98	75-125	1	20	
Selenium	mg/kg	0.38J	31.6	32.4	34.0	33.0	105	102	75-125	3	20	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824816 3824817

Parameter	Units	50396518004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Silver	mg/kg	0.26J	31.6	32.4	32.4	32.1	101	99	75-125	1	20	
Zinc	mg/kg	109	31.6	32.4	138	141	91	100	75-125	2	20	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835733
QC Batch Method: EPA 8260
Analysis Method: EPA 8260
Analysis Description: 8260 MSV 5030 Low
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824558
Matrix: Solid

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	03/24/25 10:02	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
2-Butanone (MEK)	mg/kg	<0.025	0.025	03/24/25 10:02	
2-Hexanone	mg/kg	<0.10	0.10	03/24/25 10:02	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	03/24/25 10:02	
Acetone	mg/kg	<0.10	0.10	03/24/25 10:02	
Acrolein	mg/kg	<0.10	0.10	03/24/25 10:02	
Acrylonitrile	mg/kg	<0.10	0.10	03/24/25 10:02	
Benzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Bromodichloromethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Bromoform	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Bromomethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Carbon disulfide	mg/kg	<0.010	0.010	03/24/25 10:02	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Chlorobenzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Chloroethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Chloroform	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Chloromethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Dibromochloromethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Dibromomethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	N2
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Ethyl methacrylate	mg/kg	<0.10	0.10	03/24/25 10:02	
Ethylbenzene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Iodomethane	mg/kg	<0.10	0.10	03/24/25 10:02	
Methyl methacrylate	mg/kg	<0.10	0.10	03/24/25 10:02	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	03/24/25 10:02	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

METHOD BLANK: 3824558 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	03/24/25 10:02	
Styrene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Tetrachloroethene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Toluene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	03/24/25 10:02	
Trichloroethene	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Vinyl acetate	mg/kg	<0.10	0.10	03/24/25 10:02	
Vinyl chloride	mg/kg	<0.0050	0.0050	03/24/25 10:02	
Xylene (Total)	mg/kg	<0.010	0.010	03/24/25 10:02	
4-Bromofluorobenzene (S)	%.	103	63-132	03/24/25 10:02	
Dibromofluoromethane (S)	%.	108	75-135	03/24/25 10:02	
Toluene-d8 (S)	%.	101	65-148	03/24/25 10:02	

LABORATORY CONTROL SAMPLE: 3824559

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	0.05	0.043	86	70-129	
1,1,1-Trichloroethane	mg/kg	0.05	0.040	80	67-134	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.042	84	67-122	
1,1,2-Trichloroethane	mg/kg	0.05	0.044	88	72-127	
1,1-Dichloroethane	mg/kg	0.05	0.043	86	72-121	
1,1-Dichloroethene	mg/kg	0.05	0.042	85	57-140	
1,2,3-Trichloropropane	mg/kg	0.05	0.042	85	70-124	
1,2-Dibromo-3-chloropropane	mg/kg	0.05	0.042	84	56-127	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.046	93	71-126	
1,2-Dichlorobenzene	mg/kg	0.05	0.043	85	68-120	
1,2-Dichloroethane	mg/kg	0.05	0.043	86	67-129	
1,2-Dichloropropane	mg/kg	0.05	0.045	89	71-123	
1,3-Dichlorobenzene	mg/kg	0.05	0.043	86	65-121	
1,4-Dichlorobenzene	mg/kg	0.05	0.044	88	66-122	
2-Butanone (MEK)	mg/kg	0.25	0.23	93	59-136	
2-Hexanone	mg/kg	0.25	0.20	81	62-127	
4-Methyl-2-pentanone (MIBK)	mg/kg	0.25	0.22	87	67-131	
Acetone	mg/kg	0.25	0.27	106	45-127	
Acrolein	mg/kg	1	1.2	120	42-158	
Acrylonitrile	mg/kg	0.25	0.22	88	69-127	
Benzene	mg/kg	0.05	0.042	85	69-125	
Bromodichloromethane	mg/kg	0.05	0.043	86	77-130	
Bromoform	mg/kg	0.05	0.043	85	67-128	
Bromomethane	mg/kg	0.05	0.064	129	60-156	
Carbon disulfide	mg/kg	0.05	0.046	93	47-137	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

LABORATORY CONTROL SAMPLE: 3824559

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Carbon tetrachloride	mg/kg	0.05	0.040	81	68-132	
Chlorobenzene	mg/kg	0.05	0.043	85	68-122	
Chloroethane	mg/kg	0.05	0.059	118	61-137	
Chloroform	mg/kg	0.05	0.043	85	71-124	
Chloromethane	mg/kg	0.05	0.039	79	56-131	
cis-1,2-Dichloroethene	mg/kg	0.05	0.043	87	70-123	
cis-1,3-Dichloropropene	mg/kg	0.05	0.045	91	72-136	
Dibromochloromethane	mg/kg	0.05	0.044	88	73-130	
Dibromomethane	mg/kg	0.05	0.045	90	74-123	N2
Dichlorodifluoromethane	mg/kg	0.05	0.034	67	23-127	
Ethyl methacrylate	mg/kg	0.05	<0.10	97	70-131	
Ethylbenzene	mg/kg	0.05	0.043	85	65-124	
Iodomethane	mg/kg	0.05	<0.10	103	50-137	
Methyl methacrylate	mg/kg	0.05	<0.10	96	63-137	
Methyl-tert-butyl ether	mg/kg	0.05	0.044	89	69-128	
Methylene Chloride	mg/kg	0.05	0.042	83	61-128	
Styrene	mg/kg	0.05	0.043	87	68-124	
Tetrachloroethene	mg/kg	0.05	0.045	91	62-128	
Toluene	mg/kg	0.05	0.041	83	60-122	
trans-1,2-Dichloroethene	mg/kg	0.05	0.043	87	67-124	
trans-1,3-Dichloropropene	mg/kg	0.05	0.044	89	68-136	
trans-1,4-Dichloro-2-butene	mg/kg	0.05	<0.10	85	64-134	
Trichloroethene	mg/kg	0.05	0.042	85	68-128	
Trichlorofluoromethane	mg/kg	0.05	0.052	103	57-146	
Vinyl acetate	mg/kg	0.2	0.22	108	56-181	
Vinyl chloride	mg/kg	0.05	0.045	89	52-142	
Xylene (Total)	mg/kg	0.15	0.13	85	62-122	
4-Bromofluorobenzene (S)	%.			101	63-132	
Dibromofluoromethane (S)	%.			102	75-135	
Toluene-d8 (S)	%.			102	65-148	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824560 3824561

Parameter	Units	50396648005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
1,1,1,2-Tetrachloroethane	mg/kg	ND	0.059	0.06	0.044	0.048	74	79	22-160	7	20	
1,1,1-Trichloroethane	mg/kg	ND	0.059	0.06	0.052	0.055	88	92	52-148	5	20	
1,1,2,2-Tetrachloroethane	mg/kg	ND	0.059	0.06	0.040	0.043	67	71	24-166	6	20	
1,1,2-Trichloroethane	mg/kg	ND	0.059	0.06	0.047	0.048	78	80	30-162	3	20	
1,1-Dichloroethane	mg/kg	ND	0.059	0.06	0.051	0.055	85	91	49-138	7	20	
1,1-Dichloroethene	mg/kg	ND	0.059	0.06	0.063	0.064	105	107	39-162	3	20	
1,2,3-Trichloropropane	mg/kg	ND	0.059	0.06	0.041	0.046	69	76	17-177	10	20	
1,2-Dibromo-3-chloropropane	mg/kg	ND	0.059	0.06	0.035	0.041	59	68	10-148	15	20	
1,2-Dibromoethane (EDB)	mg/kg	ND	0.059	0.06	0.046	0.047	77	79	36-141	3	20	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824560 3824561

Parameter	Units	50396648005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
1,2-Dichlorobenzene	mg/kg	ND	0.059	0.06	0.026	0.033	43	55	10-136	25	20	R1
1,2-Dichloroethane	mg/kg	ND	0.059	0.06	0.047	0.049	78	81	48-138	4	20	
1,2-Dichloropropane	mg/kg	ND	0.059	0.06	0.050	0.052	84	87	45-140	4	20	
1,3-Dichlorobenzene	mg/kg	ND	0.059	0.06	0.025	0.033	42	55	10-135	27	20	R1
1,4-Dichlorobenzene	mg/kg	ND	0.059	0.06	0.025	0.033	41	55	10-136	30	20	R1
2-Butanone (MEK)	mg/kg	ND	0.3	0.3	0.23	0.24	75	79	33-160	6	20	
2-Hexanone	mg/kg	ND	0.3	0.3	0.20	0.21	68	69	18-155	3	20	
4-Methyl-2-pentanone (MIBK)	mg/kg	ND	0.3	0.3	0.22	0.24	75	81	27-175	8	20	
Acetone	mg/kg	ND	0.3	0.3	0.28	0.29	95	95	18-159	1	20	
Acrolein	mg/kg	ND	1.2	1.2	0.69	0.57	58	47	10-155	20	20	
Acrylonitrile	mg/kg	ND	0.3	0.3	0.21	0.23	69	77	24-157	12	20	
Benzene	mg/kg	ND	0.059	0.06	0.047	0.052	79	86	48-137	9	20	
Bromodichloromethane	mg/kg	ND	0.059	0.06	0.048	0.049	80	82	32-152	4	20	
Bromoform	mg/kg	ND	0.059	0.06	0.040	0.043	67	72	10-178	9	20	
Bromomethane	mg/kg	ND	0.059	0.06	0.079	0.077	133	128	31-164	3	20	
Carbon disulfide	mg/kg	ND	0.059	0.06	0.054	0.058	90	97	23-145	8	20	
Carbon tetrachloride	mg/kg	ND	0.059	0.06	0.054	0.057	91	95	43-148	6	20	
Chlorobenzene	mg/kg	ND	0.059	0.06	0.036	0.043	61	71	28-136	17	20	
Chloroethane	mg/kg	ND	0.059	0.06	0.082	0.083	137	137	34-160	1	20	
Chloroform	mg/kg	ND	0.059	0.06	0.050	0.051	83	86	54-136	4	20	
Chloromethane	mg/kg	ND	0.059	0.06	0.055	0.053	92	88	36-145	5	20	
cis-1,2-Dichloroethene	mg/kg	ND	0.059	0.06	0.048	0.053	81	87	52-132	8	20	
cis-1,3-Dichloropropene	mg/kg	ND	0.059	0.06	0.044	0.049	74	81	22-163	9	20	
Dibromochloromethane	mg/kg	ND	0.059	0.06	0.044	0.047	74	78	18-161	6	20	
Dibromomethane	mg/kg	ND	0.059	0.06	0.046	0.048	77	80	32-147	6	20	N2
Dichlorodifluoromethane	mg/kg	ND	0.059	0.06	0.061	0.062	102	103	10-138	2	20	
Ethyl methacrylate	mg/kg	ND	0.059	0.06	<0.12	<0.12	50	43	10-167		20	
Ethylbenzene	mg/kg	ND	0.059	0.06	0.039	0.045	65	75	24-150	15	20	
Iodomethane	mg/kg	ND	0.059	0.06	<0.12	<0.12	92	95	23-142		20	
Methyl methacrylate	mg/kg	ND	0.059	0.06	<0.12	<0.12	92	96	10-177		20	
Methyl-tert-butyl ether	mg/kg	ND	0.059	0.06	0.048	0.053	81	88	57-141	9	20	
Methylene Chloride	mg/kg	ND	0.059	0.06	0.052	0.055	76	79	40-140	4	20	
Styrene	mg/kg	ND	0.059	0.06	0.033	0.040	55	66	12-138	20	20	
Tetrachloroethene	mg/kg	ND	0.059	0.06	0.054	0.061	90	101	26-159	12	20	
Toluene	mg/kg	ND	0.059	0.06	0.043	0.049	72	81	28-150	12	20	
trans-1,2-Dichloroethene	mg/kg	ND	0.059	0.06	0.048	0.057	81	94	50-134	16	20	
trans-1,3-Dichloropropene	mg/kg	ND	0.059	0.06	0.041	0.045	68	74	17-153	9	20	
trans-1,4-Dichloro-2-butene	mg/kg	ND	0.059	0.06	<0.12	<0.12	58	60	10-146		20	
Trichloroethene	mg/kg	ND	0.059	0.06	0.046	0.052	77	86	33-155	12	20	
Trichlorofluoromethane	mg/kg	ND	0.059	0.06	0.087	0.089	145	148	37-163	2	20	
Vinyl acetate	mg/kg	ND	0.23	0.25	<0.12	<0.12	31	29	10-183		20	
Vinyl chloride	mg/kg	ND	0.059	0.06	0.067	0.064	112	106	37-161	5	20	
Xylene (Total)	mg/kg	ND	0.19	0.19	0.11	0.13	63	74	25-142	17	20	
4-Bromofluorobenzene (S)	%.						101	102	63-132			
Dibromofluoromethane (S)	%.						102	102	75-135			

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824560 3824561

Parameter	Units	50396648005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Toluene-d8 (S)	%.						104	105	65-148			

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835548
QC Batch Method: EPA 5030/8260
Analysis Method: EPA 5030/8260
Analysis Description: 8260 MSV TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3823529 Matrix: Water
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	03/21/25 14:52	
1,2-Dichloroethane	mg/L	<0.050	0.050	03/21/25 14:52	
2-Butanone (MEK)	mg/L	<1.0	1.0	03/21/25 14:52	
Benzene	mg/L	<0.050	0.050	03/21/25 14:52	
Carbon tetrachloride	mg/L	<0.050	0.050	03/21/25 14:52	
Chlorobenzene	mg/L	<0.050	0.050	03/21/25 14:52	
Chloroform	mg/L	<0.050	0.050	03/21/25 14:52	
Tetrachloroethene	mg/L	<0.050	0.050	03/21/25 14:52	
Trichloroethene	mg/L	<0.050	0.050	03/21/25 14:52	
Vinyl chloride	mg/L	<0.020	0.020	03/21/25 14:52	
4-Bromofluorobenzene (S)	%.	96	79-124	03/21/25 14:52	
Dibromofluoromethane (S)	%.	103	82-128	03/21/25 14:52	
Toluene-d8 (S)	%.	103	73-122	03/21/25 14:52	

LABORATORY CONTROL SAMPLE: 3823530

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.42	84	71-130	
1,2-Dichloroethane	mg/L	0.5	0.47	94	72-123	
2-Butanone (MEK)	mg/L	2.5	2.5	99	67-135	
Benzene	mg/L	0.5	0.43	85	76-122	
Carbon tetrachloride	mg/L	0.5	0.45	91	73-127	
Chlorobenzene	mg/L	0.5	0.42	85	76-118	
Chloroform	mg/L	0.5	0.44	89	78-121	
Tetrachloroethene	mg/L	0.5	0.48	96	71-122	
Trichloroethene	mg/L	0.5	0.43	87	74-125	
Vinyl chloride	mg/L	0.5	0.52	104	55-139	
4-Bromofluorobenzene (S)	%.			104	79-124	
Dibromofluoromethane (S)	%.			100	82-128	
Toluene-d8 (S)	%.			104	73-122	

MATRIX SPIKE SAMPLE: 3823531

Parameter	Units	50396350001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.75	149	53-144	M1
1,2-Dichloroethane	mg/L	ND	0.5	0.51	102	50-138	
2-Butanone (MEK)	mg/L	ND	2.5	2.7	109	45-138	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE SAMPLE: 3823531

Parameter	Units	50396350001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.45	90	53-138	
Carbon tetrachloride	mg/L	ND	0.5	0.48	95	43-148	
Chlorobenzene	mg/L	ND	0.5	0.44	88	52-131	
Chloroform	mg/L	ND	0.5	0.45	89	54-138	
Tetrachloroethene	mg/L	ND	0.5	0.82	165	44-138	M1
Trichloroethene	mg/L	ND	0.5	0.94	187	49-140	M1
Vinyl chloride	mg/L	ND	0.5	0.53	107	41-147	
4-Bromofluorobenzene (S)	%.				104	79-124	
Dibromofluoromethane (S)	%.				19	82-128	S2
Toluene-d8 (S)	%.				102	73-122	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 836498
QC Batch Method: EPA 3546
Analysis Method: EPA 8081
Analysis Description: 8081 GCS Pesticides
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3828131 Matrix: Solid

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	03/28/25 17:41	
4,4'-DDE	ug/kg	<5.0	5.0	03/28/25 17:41	
4,4'-DDT	ug/kg	<5.0	5.0	03/28/25 17:41	
Aldrin	ug/kg	<2.5	2.5	03/28/25 17:41	
alpha-BHC	ug/kg	<2.5	2.5	03/28/25 17:41	
alpha-Chlordane	ug/kg	<2.5	2.5	03/28/25 17:41	
beta-BHC	ug/kg	<2.5	2.5	03/28/25 17:41	
Chlordane (Technical)	ug/kg	<49.8	49.8	03/28/25 17:41	
delta-BHC	ug/kg	<2.5	2.5	03/28/25 17:41	
Dieldrin	ug/kg	<5.0	5.0	03/28/25 17:41	
Endosulfan I	ug/kg	<2.5	2.5	03/28/25 17:41	
Endosulfan II	ug/kg	<5.0	5.0	03/28/25 17:41	
Endosulfan sulfate	ug/kg	<5.0	5.0	03/28/25 17:41	
Endrin	ug/kg	<5.0	5.0	03/28/25 17:41	
Endrin aldehyde	ug/kg	<5.0	5.0	03/28/25 17:41	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	03/28/25 17:41	
gamma-Chlordane	ug/kg	<2.5	2.5	03/28/25 17:41	
Heptachlor	ug/kg	<2.5	2.5	03/28/25 17:41	
Heptachlor epoxide	ug/kg	<2.5	2.5	03/28/25 17:41	
Toxaphene	ug/kg	<49.8	49.8	03/28/25 17:41	
Decachlorobiphenyl (S)	%.	75	10-139	03/28/25 17:41	

LABORATORY CONTROL SAMPLE: 3828132

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	19.8	19.7	99	34-164	
4,4'-DDE	ug/kg	19.8	19.0	96	40-156	
4,4'-DDT	ug/kg	19.8	20.6	104	29-169	
Aldrin	ug/kg	9.9	8.4	84	41-149	
alpha-BHC	ug/kg	9.9	9.0	91	41-143	
alpha-Chlordane	ug/kg	9.9	9.1	92	39-155	
beta-BHC	ug/kg	9.9	8.3	84	38-157	
delta-BHC	ug/kg	9.9	9.0	91	26-124	
Dieldrin	ug/kg	19.8	20.1	102	46-155	
Endosulfan I	ug/kg	9.9	8.8	89	39-152	
Endosulfan II	ug/kg	19.8	18.0	91	40-159	
Endosulfan sulfate	ug/kg	19.8	17.0	86	38-150	
Endrin	ug/kg	19.8	20.2	102	41-162	
Endrin aldehyde	ug/kg	19.8	18.1	91	37-196	

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

LABORATORY CONTROL SAMPLE: 3828132

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	9.9	8.9	90	39-155	
gamma-Chlordane	ug/kg	9.9	8.6	87	38-162	
Heptachlor	ug/kg	9.9	9.3	94	45-146	
Heptachlor epoxide	ug/kg	9.9	8.5	85	40-158	
Decachlorobiphenyl (S)	%.			80	10-139	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3828133 3828134

Parameter	Units	50397019004 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	19.5	19.5	18.2	19.5	93	100	28-138	7	20	
4,4'-DDE	ug/kg	ND	19.5	19.5	16.9	18.1	87	93	22-136	7	20	
4,4'-DDT	ug/kg	ND	19.5	19.5	18.0	18.9	92	97	10-164	5	20	
Aldrin	ug/kg	ND	9.8	9.8	7.5	8.2	77	84	26-137	9	20	
alpha-BHC	ug/kg	ND	9.8	9.8	8.3	8.7	86	90	31-139	5	20	
alpha-Chlordane	ug/kg	ND	9.8	9.8	8.0	8.6	82	88	30-139	7	20	
beta-BHC	ug/kg	ND	9.8	9.8	7.9	8.5	81	87	31-136	7	20	
delta-BHC	ug/kg	ND	9.8	9.8	8.8	9.3	90	95	10-149	5	20	
Dieldrin	ug/kg	ND	19.5	19.5	18.0	19.4	92	99	11-147	7	20	
Endosulfan I	ug/kg	ND	9.8	9.8	7.7	8.1	79	83	19-135	5	20	
Endosulfan II	ug/kg	ND	19.5	19.5	16.1	17.2	82	88	15-137	7	20	
Endosulfan sulfate	ug/kg	ND	19.5	19.5	15.2	15.9	78	82	18-128	5	20	
Endrin	ug/kg	ND	19.5	19.5	18.3	19.6	94	100	12-154	7	20	
Endrin aldehyde	ug/kg	ND	19.5	19.5	15.6	16.9	80	87	19-163	8	20	
gamma-BHC (Lindane)	ug/kg	ND	9.8	9.8	8.4	9.1	86	93	10-182	8	20	
gamma-Chlordane	ug/kg	ND	9.8	9.8	7.5	8.0	77	82	30-152	7	20	
Heptachlor	ug/kg	ND	9.8	9.8	8.5	9.3	87	95	22-158	9	20	
Heptachlor epoxide	ug/kg	ND	9.8	9.8	7.8	8.2	80	84	19-144	6	20	
Decachlorobiphenyl (S)	%.						59	57	10-139			

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835637
QC Batch Method: EPA 3510
Analysis Method: EPA 8081
Analysis Description: 8081 GCS TCLP Pesticides
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824307 Matrix: Water
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	03/24/25 13:28	
Endrin	mg/L	<0.00010	0.00010	03/24/25 13:28	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	03/24/25 13:28	
Heptachlor	mg/L	<0.000050	0.000050	03/24/25 13:28	
Heptachlor epoxide	mg/L	<0.000050	0.000050	03/24/25 13:28	
Methoxychlor	mg/L	<0.00050	0.00050	03/24/25 13:28	
Toxaphene	mg/L	<0.0010	0.0010	03/24/25 13:28	
Decachlorobiphenyl (S)	%.	69	10-128	03/24/25 13:28	

LABORATORY CONTROL SAMPLE: 3824308

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0019	97	41-162	
gamma-BHC (Lindane)	mg/L	0.001	0.00075	75	36-151	
Heptachlor	mg/L	0.001	0.00055	55	22-141	
Heptachlor epoxide	mg/L	0.001	0.00078	78	41-152	
Methoxychlor	mg/L	0.01	0.0095	95	45-157	
Decachlorobiphenyl (S)	%.			53	10-128	

MATRIX SPIKE SAMPLE: 3824309

Parameter	Units	50396428001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	<0.00050	0.01	0.010	102	12-162	
gamma-BHC (Lindane)	mg/L	<0.00025	0.005	0.0043	86	15-156	
Heptachlor	mg/L	<0.00025	0.005	0.0039	77	11-138	
Heptachlor epoxide	mg/L	<0.00025	0.005	0.0041	82	10-164	
Methoxychlor	mg/L	<0.0025	0.05	0.050	100	10-169	
Decachlorobiphenyl (S)	%.				61	10-128	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835607
QC Batch Method: EPA 3546
Analysis Method: EPA 8082
Analysis Description: 8082 PCB Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824078 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<100	100	03/24/25 20:24	
PCB-1221 (Aroclor 1221)	ug/kg	<100	100	03/24/25 20:24	
PCB-1232 (Aroclor 1232)	ug/kg	<100	100	03/24/25 20:24	
PCB-1242 (Aroclor 1242)	ug/kg	<100	100	03/24/25 20:24	
PCB-1248 (Aroclor 1248)	ug/kg	<100	100	03/24/25 20:24	
PCB-1254 (Aroclor 1254)	ug/kg	<100	100	03/24/25 20:24	
PCB-1260 (Aroclor 1260)	ug/kg	<100	100	03/24/25 20:24	
Tetrachloro-m-xylene (S)	%.	81	11-126	03/24/25 20:24	

LABORATORY CONTROL SAMPLE: 3824079

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	333	275	83	53-125	
PCB-1260 (Aroclor 1260)	ug/kg	333	299	90	54-134	
Tetrachloro-m-xylene (S)	%.			86	11-126	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824080 3824081

Parameter	Units	50395427001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	331	333	16500	<10000	4990	2910	10-170		20	M6
PCB-1260 (Aroclor 1260)	ug/kg	ND	331	333	<9930	<10000	2700	1660	10-156		20	M6
Tetrachloro-m-xylene (S)	%.						0	0	11-126			S4

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835780
QC Batch Method: EPA 8151
Analysis Method: EPA 8151A
Analysis Description: 8151 GCS TCLP Herbicides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3824710 Matrix: Water
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	03/28/25 16:09	
2,4-D	mg/L	<0.0010	0.0010	03/28/25 16:09	
2,4-DCAA (S)	%.	100	30-170	03/28/25 16:09	

LABORATORY CONTROL SAMPLE: 3824711

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0055	110	37-126	
2,4-D	mg/L	0.005	0.0048	96	28-126	
2,4-DCAA (S)	%.			95	30-170	

MATRIX SPIKE SAMPLE: 3824712

Parameter	Units	50396428001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0050	0.025	0.028	111	15-163	
2,4-D	mg/L	<0.0050	0.025	0.024	96	15-152	
2,4-DCAA (S)	%.				63	30-170	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835723
QC Batch Method: EPA 3546
Analysis Method: EPA 8270
Analysis Description: 8270 Solid MSSV Microwave Short Spike
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824524
Matrix: Solid

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<330	330	03/25/25 10:01	
2,4,5-Trichlorophenol	ug/kg	<330	330	03/25/25 10:01	
2,4,6-Trichlorophenol	ug/kg	<330	330	03/25/25 10:01	
2,4-Dichlorophenol	ug/kg	<330	330	03/25/25 10:01	
2,4-Dimethylphenol	ug/kg	<330	330	03/25/25 10:01	
2,4-Dinitrophenol	ug/kg	<1600	1600	03/25/25 10:01	
2,4-Dinitrotoluene	ug/kg	<330	330	03/25/25 10:01	
2,6-Dinitrotoluene	ug/kg	<330	330	03/25/25 10:01	
2-Chloronaphthalene	ug/kg	<330	330	03/25/25 10:01	
2-Chlorophenol	ug/kg	<330	330	03/25/25 10:01	
2-Methylnaphthalene	ug/kg	<330	330	03/25/25 10:01	
2-Methylphenol(o-Cresol)	ug/kg	<330	330	03/25/25 10:01	
2-Nitroaniline	ug/kg	<330	330	03/25/25 10:01	
2-Nitrophenol	ug/kg	<330	330	03/25/25 10:01	
3&4-Methylphenol(m&p Cresol)	ug/kg	<660	660	03/25/25 10:01	
3,3'-Dichlorobenzidine	ug/kg	<660	660	03/25/25 10:01	
3-Nitroaniline	ug/kg	<330	330	03/25/25 10:01	
4,6-Dinitro-2-methylphenol	ug/kg	<660	660	03/25/25 10:01	
4-Bromophenylphenyl ether	ug/kg	<330	330	03/25/25 10:01	
4-Chloro-3-methylphenol	ug/kg	<660	660	03/25/25 10:01	
4-Chloroaniline	ug/kg	<660	660	03/25/25 10:01	
4-Chlorophenylphenyl ether	ug/kg	<330	330	03/25/25 10:01	
4-Nitroaniline	ug/kg	<330	330	03/25/25 10:01	
4-Nitrophenol	ug/kg	<1600	1600	03/25/25 10:01	
Acenaphthene	ug/kg	<330	330	03/25/25 10:01	
Acenaphthylene	ug/kg	<330	330	03/25/25 10:01	
Anthracene	ug/kg	<330	330	03/25/25 10:01	
Benzo(a)anthracene	ug/kg	<330	330	03/25/25 10:01	
Benzo(a)pyrene	ug/kg	<330	330	03/25/25 10:01	
Benzo(b)fluoranthene	ug/kg	<330	330	03/25/25 10:01	
Benzo(g,h,i)perylene	ug/kg	<330	330	03/25/25 10:01	
Benzo(k)fluoranthene	ug/kg	<330	330	03/25/25 10:01	
Benzyl alcohol	ug/kg	<660	660	03/25/25 10:01	
bis(2-Chloroethoxy)methane	ug/kg	<330	330	03/25/25 10:01	
bis(2-Chloroethyl) ether	ug/kg	<330	330	03/25/25 10:01	
bis(2-Ethylhexyl)phthalate	ug/kg	<330	330	03/25/25 10:01	
bis(2chloro1methylethyl) ether	ug/kg	<330	330	03/25/25 10:01	
Butylbenzylphthalate	ug/kg	<330	330	03/25/25 10:01	
Chrysene	ug/kg	<330	330	03/25/25 10:01	
Di-n-butylphthalate	ug/kg	<330	330	03/25/25 10:01	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

METHOD BLANK: 3824524 Matrix: Solid

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<330	330	03/25/25 10:01	
Dibenz(a,h)anthracene	ug/kg	<330	330	03/25/25 10:01	
Dibenzofuran	ug/kg	<330	330	03/25/25 10:01	
Diethylphthalate	ug/kg	<330	330	03/25/25 10:01	
Dimethylphthalate	ug/kg	<330	330	03/25/25 10:01	
Fluoranthene	ug/kg	<330	330	03/25/25 10:01	
Fluorene	ug/kg	<330	330	03/25/25 10:01	
Hexachloro-1,3-butadiene	ug/kg	<330	330	03/25/25 10:01	
Hexachlorobenzene	ug/kg	<330	330	03/25/25 10:01	
Hexachlorocyclopentadiene	ug/kg	<330	330	03/25/25 10:01	
Hexachloroethane	ug/kg	<330	330	03/25/25 10:01	
Indeno(1,2,3-cd)pyrene	ug/kg	<330	330	03/25/25 10:01	
Isophorone	ug/kg	<330	330	03/25/25 10:01	
N-Nitroso-di-n-propylamine	ug/kg	<330	330	03/25/25 10:01	
N-Nitrosodiphenylamine	ug/kg	<330	330	03/25/25 10:01	
Naphthalene	ug/kg	<330	330	03/25/25 10:01	
Nitrobenzene	ug/kg	<330	330	03/25/25 10:01	
Pentachlorophenol	ug/kg	<1600	1600	03/25/25 10:01	
Phenanthrene	ug/kg	<330	330	03/25/25 10:01	
Phenol	ug/kg	<330	330	03/25/25 10:01	
Pyrene	ug/kg	<330	330	03/25/25 10:01	
2,4,6-Tribromophenol (S)	%.	101	15-140	03/25/25 10:01	
2-Fluorobiphenyl (S)	%.	85	39-97	03/25/25 10:01	
2-Fluorophenol (S)	%.	90	18-116	03/25/25 10:01	
Nitrobenzene-d5 (S)	%.	86	32-98	03/25/25 10:01	
p-Terphenyl-d14 (S)	%.	101	49-119	03/25/25 10:01	
Phenol-d5 (S)	%.	90	30-115	03/25/25 10:01	

LABORATORY CONTROL SAMPLE: 3824525

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1670	1470	88	59-121	
2-Chlorophenol	ug/kg	1670	1420	85	55-110	
2-Methylnaphthalene	ug/kg	1670	1470	88	45-116	
4-Chloro-3-methylphenol	ug/kg	1670	1540	92	60-123	
4-Nitrophenol	ug/kg	1670	<1600	83	42-141	
Acenaphthene	ug/kg	1670	1310	79	62-105	
Acenaphthylene	ug/kg	1670	1290	77	63-107	
Anthracene	ug/kg	1670	1190	72	65-109	
Benzo(a)anthracene	ug/kg	1670	1490	90	67-114	
Benzo(a)pyrene	ug/kg	1670	1440	87	62-120	
Benzo(b)fluoranthene	ug/kg	1670	1650	99	58-125	
Benzo(g,h,i)perylene	ug/kg	1670	1530	92	57-122	
Benzo(k)fluoranthene	ug/kg	1670	1420	85	64-116	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

LABORATORY CONTROL SAMPLE: 3824525

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1670	1460	87	64-116	
Dibenz(a,h)anthracene	ug/kg	1670	1500	90	58-121	
Fluoranthene	ug/kg	1670	1500	90	66-123	
Fluorene	ug/kg	1670	1490	89	66-112	
Indeno(1,2,3-cd)pyrene	ug/kg	1670	1500	90	61-121	
N-Nitroso-di-n-propylamine	ug/kg	1670	1430	86	44-106	
Naphthalene	ug/kg	1670	1290	77	56-98	
Pentachlorophenol	ug/kg	1670	<1600	69	34-128	
Phenanthrene	ug/kg	1670	1450	87	66-112	
Phenol	ug/kg	1670	1490	89	47-118	
Pyrene	ug/kg	1670	1360	82	63-116	
2,4,6-Tribromophenol (S)	%.			82	15-140	
2-Fluorobiphenyl (S)	%.			75	39-97	
2-Fluorophenol (S)	%.			84	18-116	
Nitrobenzene-d5 (S)	%.			78	32-98	
p-Terphenyl-d14 (S)	%.			84	49-119	
Phenol-d5 (S)	%.			87	30-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824526 3824527

Parameter	Units	50396205001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	11900	11400	8710	8350	73	73	14-127	4	20	
2-Chlorophenol	ug/kg	ND	11800	11400	9670	8950	82	78	10-130	8	20	
2-Methylnaphthalene	ug/kg	ND	11800	11400	10200	9460	86	83	19-131	7	20	
4-Chloro-3-methylphenol	ug/kg	ND	11800	11400	10400	9510	88	83	18-139	9	20	
4-Nitrophenol	ug/kg	ND	11800	11400	<11400	<10900	75	75	10-142		20	
Acenaphthene	ug/kg	ND	11800	11400	8860	8480	75	74	13-133	4	20	
Acenaphthylene	ug/kg	ND	11800	11400	8740	8070	74	71	17-129	8	20	
Anthracene	ug/kg	ND	11800	11400	7850	7390	66	65	18-130	6	20	
Benzo(a)anthracene	ug/kg	ND	11800	11400	9120	8820	77	77	19-135	3	20	
Benzo(a)pyrene	ug/kg	ND	11800	11400	8970	8560	76	75	19-133	5	20	
Benzo(b)fluoranthene	ug/kg	ND	11800	11400	9270	8610	78	75	11-142	7	20	
Benzo(g,h,i)perylene	ug/kg	ND	11800	11400	9160	8690	77	76	12-134	5	20	
Benzo(k)fluoranthene	ug/kg	ND	11800	11400	9760	9380	82	82	19-132	4	20	
Chrysene	ug/kg	ND	11800	11400	9490	9420	80	83	14-140	1	20	
Dibenz(a,h)anthracene	ug/kg	ND	11800	11400	9080	8590	77	75	10-146	6	20	
Fluoranthene	ug/kg	ND	11800	11400	9600	9390	81	82	12-149	2	20	
Fluorene	ug/kg	ND	11800	11400	10000	9600	85	84	17-134	5	20	
Indeno(1,2,3-cd)pyrene	ug/kg	ND	11800	11400	9170	8550	77	75	12-138	7	20	
N-Nitroso-di-n-propylamine	ug/kg	ND	11800	11400	9660	9120	82	80	17-118	6	20	
Naphthalene	ug/kg	ND	11800	11400	8790	8240	74	72	17-124	7	20	
Pentachlorophenol	ug/kg	ND	11800	11400	<11400	<10900	41	36	10-125		20	
Phenanthrene	ug/kg	ND	11800	11400	9450	8870	80	78	14-140	6	20	
Phenol	ug/kg	ND	11800	11400	9980	9420	84	83	15-128	6	20	P1

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824526 3824527

Parameter	Units	50396205001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Pyrene	ug/kg	ND	11800	11400	9290	8950	79	79	14-147	4	20	
2,4,6-Tribromophenol (S)	%.						68	64	15-140			
2-Fluorobiphenyl (S)	%.						68	66	39-97			
2-Fluorophenol (S)	%.						80	78	18-116			
Nitrobenzene-d5 (S)	%.						73	68	32-98			
p-Terphenyl-d14 (S)	%.						76	74	49-119			
Phenol-d5 (S)	%.						82	80	30-115			

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REPORT OF LABORATORY ANALYSIS

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4171 40th St. SE
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 836186
QC Batch Method: EPA 3510
Analysis Method: EPA 8270
Analysis Description: 8270 TCLP MSSV
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3826431 Matrix: Water

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	03/29/25 20:47	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	03/29/25 20:47	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	03/29/25 20:47	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	03/29/25 20:47	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	03/29/25 20:47	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	03/29/25 20:47	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	03/29/25 20:47	
Hexachlorobenzene	mg/L	<0.010	0.010	03/29/25 20:47	
Hexachloroethane	mg/L	<0.010	0.010	03/29/25 20:47	
Nitrobenzene	mg/L	<0.010	0.010	03/29/25 20:47	
Pentachlorophenol	mg/L	<0.050	0.050	03/29/25 20:47	
Pyridine	mg/L	<0.010	0.010	03/29/25 20:47	
2,4,6-Tribromophenol (S)	%.	92	49-147	03/29/25 20:47	
2-Fluorobiphenyl (S)	%.	59	29-84	03/29/25 20:47	
2-Fluorophenol (S)	%.	46	25-76	03/29/25 20:47	
Nitrobenzene-d5 (S)	%.	74	41-102	03/29/25 20:47	
p-Terphenyl-d14 (S)	%.	107	54-119	03/29/25 20:47	
Phenol-d5 (S)	%.	32	18-51	03/29/25 20:47	

LABORATORY CONTROL SAMPLE: 3826432

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.031	63	15-85	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	94	47-116	
2,4,6-Trichlorophenol	mg/L	0.05	0.046	92	48-115	
2,4-Dinitrotoluene	mg/L	0.05	0.048	96	44-112	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.038	76	39-98	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.070	70	35-92	
Hexachloro-1,3-butadiene	mg/L	0.05	0.031	62	10-81	
Hexachlorobenzene	mg/L	0.05	0.034	69	35-98	
Hexachloroethane	mg/L	0.05	0.031	61	10-83	
Nitrobenzene	mg/L	0.05	0.042	84	45-110	
Pentachlorophenol	mg/L	0.05	<0.050	42	24-127	
Pyridine	mg/L	0.05	0.026	52	10-83	
2,4,6-Tribromophenol (S)	%.			91	49-147	
2-Fluorobiphenyl (S)	%.			71	29-84	
2-Fluorophenol (S)	%.			55	25-76	
Nitrobenzene-d5 (S)	%.			84	41-102	
p-Terphenyl-d14 (S)	%.			106	54-119	

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

LABORATORY CONTROL SAMPLE: 3826432

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			38	18-51	

MATRIX SPIKE SAMPLE: 3826433

Parameter	Units	50396350001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.37	74	17-87	
2,4,5-Trichlorophenol	mg/L	ND	0.5	0.59	117	43-126	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.53	106	43-123	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.46	92	48-106	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.43	86	38-100	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.80	80	31-98	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.37	74	10-90	
Hexachlorobenzene	mg/L	ND	0.5	0.33	66	32-94	
Hexachloroethane	mg/L	ND	0.5	0.36	72	10-88	
Nitrobenzene	mg/L	ND	0.5	0.38	76	46-105	
Pentachlorophenol	mg/L	ND	0.5	0.53	105	20-141	
Pyridine	mg/L	ND	0.5	0.24	48	10-85	
2,4,6-Tribromophenol (S)	%.				105	49-147	
2-Fluorobiphenyl (S)	%.				60	29-84	
2-Fluorophenol (S)	%.				56	25-76	
Nitrobenzene-d5 (S)	%.				71	41-102	
p-Terphenyl-d14 (S)	%.				88	54-119	
Phenol-d5 (S)	%.				42	18-51	

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Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835235
QC Batch Method: SM 2540G
Analysis Method: SM 2540G
Analysis Description: Dry Weight/Percent Moisture
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

SAMPLE DUPLICATE: 3821647

Parameter	Units	50396428001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	96.9	96.9	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch:	835236	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

SAMPLE DUPLICATE: 3821649

Parameter	Units	50396428001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	3.1	3.1	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 836402
QC Batch Method: EPA 9038
Analysis Method: EPA 9038
Analysis Description: 9038 S2ulfate Soil
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3827628 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<100	100	03/28/25 09:23	N2

LABORATORY CONTROL SAMPLE: 3827629

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	205	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3827630 3827631

Parameter	Units	50396474001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	1280	2590	2590	4340	4460	117	121	90-110	3	20	M3,N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch:	836297	Analysis Method:	EPA 9045
QC Batch Method:	EPA 9045	Analysis Description:	9045 pH
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

SAMPLE DUPLICATE: 3827146

Parameter	Units	50396786001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.1	7.0	1	2	H3

SAMPLE DUPLICATE: 3827147

Parameter	Units	50396802001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	5.3	5.3	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835709
QC Batch Method: EPA 351.2
Analysis Method: EPA 351.2
Analysis Description: 351.2 TKN
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824444 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<48.8	48.8	03/27/25 11:38	N2

LABORATORY CONTROL SAMPLE: 3824445

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	465	457	98	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824446 3824447

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	69400	14600	15700	91900	76500	154	45	90-110	18	20	M3,N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835972
QC Batch Method: EPA 353.2
Analysis Method: EPA 353.2
Analysis Description: 353.2 Nitrate + Nitrite
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50396428001

METHOD BLANK: 3825405 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	03/26/25 08:51	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	03/26/25 08:51	N2

LABORATORY CONTROL SAMPLE: 3825406

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.3	103	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.9	20.7	104	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3825407 3825408

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	<1610	3210	3210	3270	3270	101	101	90-110	0	20	N2
Nitrogen, NO2 plus NO3	mg/kg	<1610	6440	6440	6610	6610	102	102	90-110	0	20	N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 836070
QC Batch Method: SM 4500-NH3 B
Analysis Method: SM 4500-NH3 G
Analysis Description: 4500 Ammonia, Distilled
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3826000 Matrix: Solid

Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	03/26/25 14:59	N2

LABORATORY CONTROL SAMPLE: 3826001

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	20	19.7	99	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3826002 3826003

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	17400	3860	6180	25600	40600	212	375	90-110	45	20	M3,N2

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835782
QC Batch Method: SM 4500-Cl-E
Analysis Method: SM 4500-Cl-E
Analysis Description: 4500 Chloride
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3824717 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.7	99.7	03/25/25 10:29	N2

LABORATORY CONTROL SAMPLE: 3824718

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	216	108	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3824719 3824720

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<16100	32100	32100	42200	41100	114	110	90-110	3	20	M0,N2

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QUALITY CONTROL DATA

Project: Biosolids-Spring
Pace Project No.: 50396428

QC Batch: 835915
QC Batch Method: SM 4500-CN-C
Analysis Method: SM 4500-CN-E
Analysis Description: 4500CNE Cyanide
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50396428001

METHOD BLANK: 3825185 Matrix: Solid
Associated Lab Samples: 50396428001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Cyanide	mg/kg	<0.46	0.46	03/27/25 14:44	N2

LABORATORY CONTROL SAMPLE: 3825186

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	4.9	4.8	98	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3825187 3825188

Parameter	Units	50396428001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	<15.6	151	158	137	142	90	91	90-110	4	20	N2

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QUALIFIERS

Project:	Biosolids-Spring
Pace Project No.:	50396428

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H3	Sample was received or analysis requested beyond the recognized method holding time.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
M6	Matrix spike and Matrix spike duplicate recovery not evaluated against control limits due to sample dilution.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.
S2	Surrogate recovery outside laboratory control limits due to matrix interferences (confirmed by similar results from sample re-analysis).
S4	Surrogate recovery not evaluated against control limits due to sample dilution.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids-Spring
Pace Project No.: 50396428

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50396428001	**DIGESTER #3**	EPA 3546	836498	EPA 8081	836575
50396428001	**DIGESTER #3**	EPA 3510	835637	EPA 8081	835750
50396428001	**DIGESTER #3**	EPA 3546	835607	EPA 8082	835736
50396428001	**DIGESTER #3**	EPA 8151	835780	EPA 8151A	836157
50396428001	**DIGESTER #3**	EPA 3050	835177	EPA 6010	836110
50396428001	**DIGESTER #3**	EPA 3010	835549	EPA 6010	836350
50396428001	**DIGESTER #3**	EPA 3050B	835811	EPA 6020	836240
50396428001	**DIGESTER #3**	EPA 7470	835468	EPA 7470	835674
50396428001	**DIGESTER #3**	EPA 7471	836325	EPA 7471	836739
50396428001	**DIGESTER #3**	EPA 3546	835723	EPA 8270	835911
50396428001	**DIGESTER #3**	EPA 3510	836186	EPA 8270	836332
50396428001	**DIGESTER #3**	EPA 8260	835733		
50396428001	**DIGESTER #3**	EPA 5030/8260	835548		
50396428001	**DIGESTER #3**	SM 2540G	835235		
50396428001	**DIGESTER #3**	SM 2540G	835236		
50396428001	**DIGESTER #3**	EPA 9038	836402	EPA 9038	836477
50396428001	**DIGESTER #3**	EPA 9045	836297		
50396428001	**DIGESTER #3**	EPA 351.2	835709	EPA 351.2	836210
50396428001	**DIGESTER #3**	EPA 353.2	835972	EPA 353.2	835974
50396428001	**DIGESTER #3**	SM 4500-NH3 B	836070	SM 4500-NH3 G	836130
50396428001	**DIGESTER #3**	SM 4500-Cl-E	835782	SM 4500-Cl-E	835783
50396428001	**DIGESTER #3**	SM 4500-CN-C	835915	SM 4500-CN-E	836415

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY- Affix Workorder/Login Label Here

WO#: 50396428

50396428

Company Name: City of Cadillac
Street Address: 1121 Plett Rd
Cadillac, MI 49601
Customer Project #:
Project Name: Biosolids - Spring
Site Collection Info/Facility ID (as applicable):

Contact/Report To: Cindy Tomaszewski
Phone #: 231-775-2368
E-Mail: ctomaszewski@cadillac-mi.net
Cc E-Mail:
Invoice To: Cindy Tomaszewski
Invoice E-Mail: ctomaszewski@cadillac-mi.net
Purchase Order # (if applicable):
Quote #:

Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [X] ET
County / State origin of sample(s): Michigan

Data Deliverables:
[ ] Level II [ ] Level III [ ] Level IV
[ ] EQUIS
[ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: NPDES
Reportable [ ] Yes [ ] No

Rush (Pre-approval required):
[ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ______

DW PWSID # or WW Permit # as applicable:

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [X] No
Analysis:

Specify Container Size **: 10, 10, 10, 10
Identify Container Preservative Type***: 1, 1, 1, 1

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only
Proj. Mgr: Brian Hall
AcctNum / Client ID:
Table #:
Profile / Template: 8311
Prelog / Bottle Ord. ID: EZ 3084205

Preservation non-conformance identified for sample.

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW), Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Analysis Requested

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	8260 MSV 5030 VOC	8270 SVOC; 8081 Pest; 8082 PCB	Metals; Wet Chem	TCLP - VOC/SV/Met/Pest/Herb	Sample Comment
DIGESTER #3	BS	G	—	—	3-17-25	1314	4	-	-	X	X	X	X	

Additional Instructions from Pace®:
Wet Chem: TKN,NH3,CN,Cl,NO2/NO3,SO4,pH,%Solid,TVS
17 Metals: 6010/6020/7471

Collected By:
(Printed Name) Taysha Laczko
Signature: T. Laczko

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
T. Laczko / City of Cadillac	3-17-25 / 1330	Ship via UPS Ground	
		Lynn Reyel / Pace	3-18-25 10:25

Tracking Number:

Delivered by: [ ] In-Person [ ] Courier
[ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 3/18 1329	Evaluated By: JAS	WO#:50396428	
Client: City of Cadillac	PM: BJH	PM: BJH	Due Date: 04/01/25
Lab Notified of Rush or Short Holds: YES NO NA		CLIENT: GR-Cadillac	

Project Received Via: FedEx UPS Client Pace Courier Other: ______				Comments:
Custody Seal Present and Intact:	YES	NO	NA	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA	
Short Hold Present (≤ 48 Hours):	YES	NO		
Sample Received in Hold:	YES	NO		
Custody Signature Present:	YES	NO		
Collector Signature Present:	YES	NO		
Sample Collected Today and On Ice:	YES	NO	N/A	
IR Gun #: 350 351 **Therm #:** 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)			
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 3.3/3.0 °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: ______ °C			
Temp Blank Received:	YES	NO		
Sample Label Matches COC (ID/Date/Time):	YES	NO		
Container Intact:	YES	NO		
Correct Container:	YES	NO		
Sufficient Volume:	YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. **pH Strip Lot #:** ______ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	N/A	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** ______ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	NO	N/A	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A	
Trip Blank Received: HCl MeOH Other: ______	YES	NO	ON HOLD	

Comments:	3. Cooler Temp. Upon Receipt: ______ °C
	4. Cooler Temp. Upon Receipt: ______ °C
	Non-Conformance Form Required: YES NO

F-GR-C-007-rev.08, 19Jul2023